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Capsule Protocol Specification

An Open Reference Architecture for Structured AI Agent Decision Records

Response to Request for Information

Security Considerations for Artificial Intelligence Agents

National Institute of Standards and Technology · Center for AI Standards and Innovation

This submission describes the Capsule Protocol Specification (CPS), a structured record format for cryptographically sealed AI agent decision records. CPS defines six mandatory sections per record, cross-language canonical serialization, and hash chain integrity verification. The protocol is open-source (Apache-2.0) with perpetual, royalty-free patent grants.

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Apache-2.0 · Royalty-free patent grant

At a Glance

Every day, AI agents approve financial transactions, modify production infrastructure, and recommend clinical interventions. When those decisions are questioned by regulators, auditors, or incident response teams, **no standard defines what evidence should exist.**

When structured decision records exist, regulators can verify AI decisions independently. Incident response starts from sealed evidence, not reconstructed timelines. Compliance becomes auditable, not self-reported. A common record format enables alignment across industries, agencies, and borders.

We recommend NIST consider defining a minimum required structure for AI agent decision records across autonomous systems. To demonstrate that such a standard is feasible, this response includes a complete, working protocol specification built on a single principle:

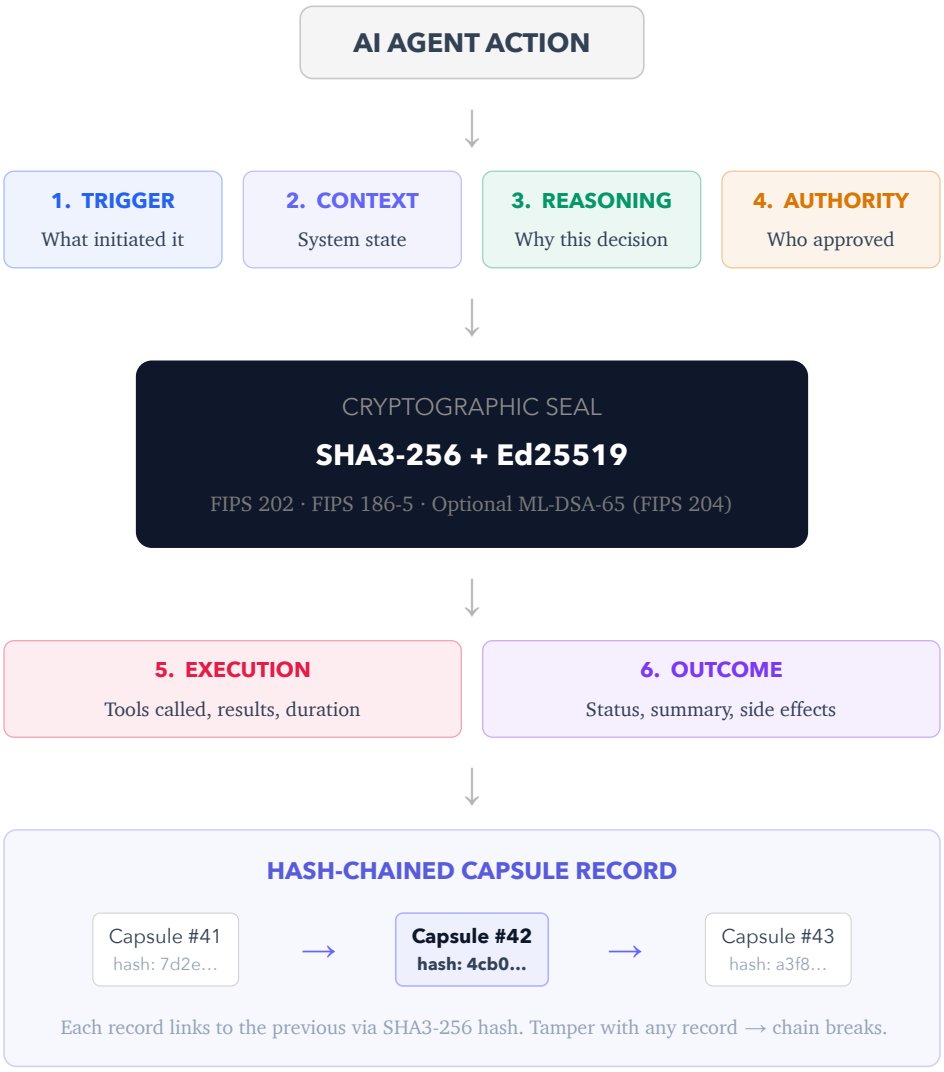
For every action, there exists a Capsule.

A Capsule is a cryptographically sealed record of a single AI agent action: what happened, why the AI decided to act, who authorized it, and what the outcome was. Six mandatory sections. Sealed with SHA3-256 and Ed25519 at the moment of action. Linked in a hash chain where tampering with any record invalidates every record that follows.

The Capsule Protocol Specification (CPS), two reference implementations (Python and TypeScript), and a 15-vector conformance suite are openly available under Apache-2.0 with royalty-free patent grants, offered to inform NIST's standards development work.

Capsule Record Lifecycle

How a single AI agent action becomes a tamper-evident, verifiable record



Reasoning is captured before execution. The seal is applied before anyone knows the record will matter.

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Introduction

AI systems are beginning to make consequential decisions autonomously, yet no widely adopted standard defines what evidence must exist to explain those decisions after the fact.

Without contemporaneous decision records, organizations must reconstruct AI behavior from incomplete logs and system artifacts after incidents occur.

I am the founder of Quantum Pipes Technologies, an open-source AI infrastructure company that has designed, built, and shipped a production-ready protocol for cryptographic accountability of autonomous AI agent actions. I am responding to this RFI not from a policy or theoretical perspective, but as an engineer who has designed, tested, and deployed the exact type of system this RFI asks about.

> **What We Contribute**
>
> The **Capsule Protocol Specification (CPS)**: an open protocol that defines how AI agent actions are recorded, sealed, verified, and chained into tamper-evident audit trails. CPS is a protocol specification, not a product. Any organization can implement it independently using the specification and conformance suite. We offer it as a concrete, production-tested starting point for what could become a NIST standard.

To our knowledge, no existing open standard defines a structured, cryptographically sealed record format specifically for AI agent decision records with pre-execution reasoning capture. The CPS specification is offered as an open reference design intended to support experimentation, independent implementations, and potential standards development.

Protocol and Reference Implementations

The complete specification, conformance suite, documentation, test vectors, and two reference implementations are openly available:

`github.com/quantumpipes/capsule`

- **Open source** under Apache-2.0 with perpetual, royalty-free patent grants. U.S. provisional patent filed February 2026 with explicit grant: free to implement for any purpose, including government systems and federally-funded research, with no licensing fees, royalties, or restrictions.
- **Cross-language by design:** Two conformant reference implementations (Python and TypeScript), both passing all 15 golden test vectors with 100% test coverage. The protocol is language-agnostic; the reference implementations prove cross-language interoperability.
- **600+ automated tests with 100% coverage** across critical, property-based, golden, conformance, unit, and integration categories, in both Python and TypeScript.
- **Protocol-first repository structure:** `spec/` (normative specification), `conformance/` (golden test vectors), `reference/python/` and `reference/typescript/` (implementations). Structured like IETF and W3C protocol repositories.
- **Air-gapped deployment** validated on workstation-class GPU hardware (HP Z and ZGX series). Zero external network dependencies.

The CPS v1.0 specification, published openly at `github.com/quantumpipes/capsule/tree/main/spec`, defines the complete protocol: record structure, serialization rules, sealing procedures, hash chain linking, and 15 golden test vectors for cross-implementation conformance. Any implementation that passes these vectors produces byte-identical hashes for identical Capsules.

These test vectors are available to support independent implementations, verification tools, and potential standards development activities, including future NIST conformance testing efforts.

What Is a Capsule?

A **Capsule** is a cryptographically sealed record of a single AI agent action. Each captures six mandatory sections, is hashed with SHA3-256, signed with Ed25519 (optional ML-DSA-65 post-quantum dual signature), and linked in a hash chain.

1. Trigger

What initiated the action. Type (user request, scheduled, system, agent), source identifier, timestamp, request description, user ID, and correlation ID for distributed tracing.

2. Context

System state at the time of action. Agent identifier, session ID for conversation tracking, and environment variables (model, workspace, hardware configuration).

3. Reasoning

The AI's decision process, captured **before execution begins**. Structured options with pros/cons and feasibility scores, selected option with rationale, confidence (0.0 to 1.0), model, and prompt hash.

4. Authority

Authorization chain. Was this autonomous, human-approved, policy-driven, or escalated? Records approver identity, policy reference, multi-level approval chain, escalation reason, and confidence threshold.

5. Execution

What actually happened. Each tool call with arguments, result, success/failure status, and duration. Total duration and resources consumed (CPU, API calls, tokens).

6. Outcome

The result. Status (success, failure, partial, blocked), human-readable summary, error details, side effects logged, and performance metrics (tokens in/out, cost, latency).

Cryptographic Seal

FIPS 202 · FIPS 186-5 · FIPS 204

Capsule content → canonical JSON → SHA3-256 hash → Ed25519 signature (required) → optional ML-DSA-65 post-quantum dual signature. Sealed at the moment of action, before anyone knows the record will matter.

Hash Chain

Tamper-evident ordering

Each Capsule records the SHA3-256 hash of the previous, forming a sequential chain. Modify, delete, or insert any record → every subsequent hash changes. No consensus mechanism or distributed ledger required.

Six sections. One truth. Tamper-evident forever.

THE FOUNDATIONAL AXIOM

$$\forall \text{ action} : \exists \text{ capsule}$$

For every action, there exists a Capsule.

1. Security Threats, Risks, and Vulnerabilities

1(a): Unique security threats affecting AI agent systems PRIORITY

The most consequential security threat unique to AI agent systems is **the inability to reconstruct what happened and why after an incident**. Traditional software systems produce logs. AI agent systems make *decisions*: choosing between options, evaluating risk, selecting tools, and acting on the world. When something goes wrong, the question is not "what did the system do?" but "why did it decide to do that?"

Traditional audit logs (syslog, database trails, application-level logging) are fundamentally insufficient for AI agent accountability:

Limitation	Why It Matters for AI Agents
Mutable	Log entries can be modified or deleted without detection. An operator under investigation can alter the record.
Actions only	A log records that a tool was called. It does not record *why* the AI chose that tool, what alternatives it considered, or who authorized the action.
No crypto integrity	No mechanism to prove the log is complete and unmodified. An auditor must trust the operator, precisely when the operator may have incentive to present a favorable record.
Post-hoc	Organizations reconstruct what happened from fragments. Contemporaneous evidence, sealed cryptographically at the moment of action, carries fundamentally stronger evidentiary weight.

CPS addresses each of these:

- **Immutability**: Each Capsule is sealed with SHA3-256 + Ed25519 at the moment of creation. Capsules are linked in a hash chain; tampering with any record invalidates every record that follows.
- **Structured reasoning capture**: The Reasoning section records the AI's analysis, options considered, selected option, confidence level, and model, all captured *before* execution.
- **Authority tracking**: The Authority section records whether the action was autonomous, human-approved, policy-driven, or escalated.

- **Contemporaneous sealing**: The cryptographic seal is applied at the moment of action. The record exists before anyone asks about it.

> **Formal Security Properties**
>
> CPS provides five verifiable properties: (1) **Integrity**: sealed content cannot be modified without detection. (2) **Authenticity**: records are signed by a known key. (3) **Non-repudiation**: the signer cannot deny having signed. (4) **Chain integrity**: deletion, insertion, or reordering is detectable. (5) **Temporal ordering**: capsules in a chain have a verifiable sequence. CPS explicitly does *not* guarantee confidentiality (records are plaintext), truthfulness (the AI could record false reasoning), or clock accuracy (timestamps are self-reported).

1(d): How have threats evolved? PRIORITY

As AI agents gain broader autonomy and higher-stakes decision authority (financial transactions, infrastructure management, clinical support), the accountability gap widens. An AI agent that schedules a meeting has low consequences if it errs. An AI agent that approves a $2M purchase order, modifies production infrastructure, or recommends a clinical intervention demands provable, tamper-evident records.

The scale is already significant. The CyberArk 2025 Identity Security Landscape report estimates 82 non-human identities for every human user in enterprise environments, the majority operating outside security team visibility. Agent task complexity and autonomy are increasing rapidly, and the OWASP Top 10 for Agentic Applications (December 2025) identifies ten distinct risk categories specific to autonomous AI systems.

> **The Trajectory**
>
> AI agents are being given more authority, operating with less human oversight, and making decisions with larger consequences. The security infrastructure for accountability must advance at the same pace as the autonomy. It currently does not.

2. Security Practices

2(a): Technical controls for secure development and deployment

PRIORITY

The following controls are implemented and tested in the `qp-capsule` reference implementation:

Cryptographic Sealing

Every Capsule sealed with **SHA3-256** (FIPS 202, resistant to length extension attacks) + **Ed25519** (RFC 8032 / FIPS 186-5). Optional **ML-DSA-65** (FIPS 204) post-quantum dual signature. Ed25519 is mandatory. An unsigned record is not evidence.

Hash Chain Integrity

Each Capsule links to the previous via hash, requiring no consensus mechanism or distributed ledger. Sequential, gapless numbering. Modification, deletion, or insertion of any record is detectable through hash recomputation.

Conformance Testing

Byte-level serialization rules ensure any conformant implementation produces identical hashes. 15 golden test vectors define the conformance contract.

Air-Gapped Operation

Zero external network dependencies. Runs offline. Designed for SCIF, clinical, and financial environments where data cannot leave the perimeter.

Kill Switch: Cannot Be Disabled

Emergency termination of any AI agent. **SOFT**: complete current action, then stop. **HARD**: immediate termination. Cannot be disabled; every activation creates its own sealed Capsule.

Scope: What CPS Does and Does Not Address

CPS addresses the accountability layer of AI agent security. It does not replace model-level controls (prompt injection robustness, safety alignment), system-level access controls (authorization scope, least privilege), or identity infrastructure (agent authentication, delegation chains). These are complementary: a prompt injection attack that succeeds despite preventive controls produces a sealed, tamper-evident Capsule that enables forensic reconstruction. Prevention reduces incidents; accountability ensures incidents are reconstructable.

2(e): Relevant cybersecurity frameworks PRIORITY

Framework	Alignment	CPS Mechanism
SP 800-53 r5	AU-10 Non-repudiation via Ed25519, AU-9 Audit protection via hash chain, AU-11 Retention	Ed25519 + hash chain
AI RMF 1.0	MEASURE 2.6, MANAGE 2.2, GOVERN 1.5	Capsule analytics
SP 800-218A	Secure development profile	Sealing in execution loop
ISO 42001	AI management system requirements	Capsule trails as evidence
FIPS 202	SHA3-256 content integrity	Canonical hash computation
FIPS 186-5	Ed25519 digital signatures	Mandatory per-Capsule signing
FIPS 204	ML-DSA-65 post-quantum signatures	Optional dual signature
SP 800-207	Zero Trust continuous verification, assume breach	Per-action sealing + chain verification
EU AI Act	Articles 12 (record-keeping), 13 (transparency), 14 (human oversight)	Six sections + authority chain
OWASP Agentic Top 10	Accountability across all 10 agent risk categories	Capsule trails as forensic evidence

3. Assessing the Security of AI Agent Systems

3(a): Methods during development to anticipate security threats

PRIORITY

The Critical Innovation: Pre-Execution Reasoning Capture

Most audit systems record *what* happened. The Capsule Reasoning section records *why* the AI decided to act, including the analysis, options considered, selected option, rationale, confidence score (0.0–1.0), the model used, and a SHA3-256 hash of the full prompt. This reasoning is captured **before execution begins**.

The temporal ordering is enforced by the data model: Reasoning is populated, Authority determines approval, and only then does Execution record tool calls.

What This Means for Incident Investigation

- Investigators see exactly what the AI considered and why it chose a particular course
- Reasoning cannot be fabricated after the fact because it was cryptographically sealed before execution
- Confidence scores provide quantitative evidence of the AI's certainty at the moment of decision
- Prompt hashes allow verification that reasoning was generated from a specific input
- Implementations may redact or hash sensitive reasoning inputs while preserving structural evidence

Test Suite as Security Assessment

- **Critical**: Capsule integrity, Kill Switch reliability (paths that must never fail)
- **Antifragile**: Property-based invariants ensuring the system holds under inputs we haven't imagined
- **Golden**: Cross-language conformance via 15 test vectors defining byte-identical canonical JSON and hashes

3(b): How could the security of a particular AI agent system be assessed? PRIORITY

An organization assessing AI agent accountability should verify five properties:

1. Is every agent action recorded?
Not sampled, not batched. Every action. The CPS axiom: $\forall\, action : \exists\, capsule$.

2. Is reasoning captured before execution?
If reasoning is reconstructed after the fact, it is not contemporaneous evidence.

3. Are records cryptographically sealed at the moment of creation?
If records are signed later, the gap between action and sealing is a window for tampering.

4. Can records be verified independently?
An auditor must verify hash chains and signatures without cooperation from the entity being audited.

5. Are there cross-implementation conformance tests?
If verification tools come only from the operator, the operator can produce tools that validate fabricated records.

4. Deployment Environments

4(a): How should deployment environments be constrained? PRIORITY

AI agent systems handling sensitive operations should be constrained to environments where the audit trail cannot be exfiltrated, modified, or bypassed:

- **Zero external network requests.** The entire platform runs on local infrastructure. No CDN, no analytics, no telemetry, no phone-home.
- **Air-gap ready.** Fully disconnected operation. Cryptographic keys generated locally. Signature verification requires no external service.
- **Workstation-class hardware.** Validated on HP Z-series and ZGX GPU workstations with full air-gapped operation, and on Apple Silicon (M-series) macOS systems. Designed for deployment on enterprise workstation infrastructure without modification.

4(b): How could environments be modified to mitigate security risks? PRIORITY

The Kill Switch is the primary environment modification mechanism. When an AI agent exhibits unexpected behavior, any authorized user can issue a SOFT kill (complete current action, then stop) or HARD kill (immediate termination). The Kill Switch creates its own Capsule record and cannot be disabled, overridden, or circumvented.

This addresses the “undo” problem: while external actions cannot be reversed (a sent email cannot be unsent), further actions can be immediately halted with a cryptographic record of when and why.

4(d): Methods for monitoring deployment environments PRIORITY

The Capsule chain itself is the monitoring mechanism. Because every action produces a structured, sealed record, monitoring reduces to queries over a known schema rather than log parsing:

Chain Integrity

Periodically recompute hashes to confirm no records have been modified, deleted, or inserted. A single tampered record invalidates every record that follows.

Detects: Record tampering, deletion, insertion, incomplete chains.

Reasoning Analysis

Review Reasoning sections for anomalous confidence patterns, unexpected option selection, or declining confidence over time. Flag reasoning that contradicts stated policy.

Detects: Model degradation, prompt injection influence, specification gaming.

Authority Auditing

Verify actions requiring human approval actually received it. Flag autonomous actions in domains that should require escalation.

Detects: Authorization bypass, policy drift, privilege escalation.

Execution Monitoring

Track tool call patterns, resource consumption, duration, and side effects to detect behavioral drift from established baselines.

Detects: Anomalous tool usage, resource abuse, unexpected side effects.

Critically, these monitoring capabilities emerge from the record structure itself. No additional instrumentation is required. CPS includes three built-in grouping mechanisms: `session_id` groups records within a conversation or workflow, `parent_id` links hierarchical actions (e.g., WORKFLOW → AGENT → TOOL), and `correlation_id` traces operations across distributed services. Together with the hash chain, these transform monitoring from “search through logs” to “query a structured, verified database.”

5. Additional Considerations

5(a): What would aid rapid adoption of security practices?

An open, cross-language specification for structured AI agent decision records: not just event logs, but records that capture *why* the AI decided to act.

CPS v1.0 Defines

- 6-section record schema
- Canonical JSON serialization
- SHA3-256 + Ed25519 sealing
- Hash chain linking rules
- Golden test vectors
- Implementation checklist
- Kill switch & verification procedures

Open by Design

- Apache-2.0 licensed
- 600+ tests, 100% coverage
- Reference Python & TypeScript
- Cross-language golden test vectors
- U.S. patent with royalty-free grants
- Free for government systems
- Free for federally-funded research

Per-action sealing introduces computational overhead (each Capsule requires a hash computation and Ed25519 signature), and chains grow linearly with agent activity. We consider this an acceptable cost. The alternative of reconstructing accountability from mutable, unstructured logs after an incident is not a viable security posture for autonomous AI systems.

5(b): Where is government collaboration most urgent?

Defining what an AI agent audit record should contain. Many organizations are building logging and monitoring for AI agents. Without a structured standard for what information must be captured, particularly the AI's reasoning process before execution, the industry will converge on minimal, unstructured event logs that record *that* things happened but not *why*.

The distinction matters. An event log that says "agent called tool X at time T" is operationally useful but weak as evidence. A structured record that says "the agent analyzed the situation, considered three options, selected option B with 0.87 confidence because of factors X, Y, Z, and the action was authorized by policy P," sealed cryptographically at the moment of decision, is contemporaneous evidence. Under the Federal Rules of Evidence (Rule 803(6), Records of Regularly Conducted Activity), records created at or near the time of the event, by a person or process with knowledge, kept in the course of a regularly conducted activity, carry stronger evidentiary weight than reconstructed accounts. CPS records satisfy these criteria by design.

The NCCoE concept paper on AI Agent Identity and Authorization (February 2026) identifies four scope areas, including logging and transparency. CPS is designed to serve exactly that function, and we welcome the opportunity to contribute to that effort.

CPS can also serve as a reference model for defining the minimum required structure of AI agent audit records, independent of any specific implementation. Even if organizations adopt alternative storage or cryptographic mechanisms, the six-section record structure provides a practical baseline for decision accountability.

Proposed Standards Direction

NIST is uniquely positioned to define the minimum required structure for AI agent decision records across autonomous systems.

Such a standard could specify:

- The **required fields** for AI decision records
- **Temporal ordering requirements** between reasoning, authorization, and execution
- **Cryptographic integrity mechanisms** for tamper detection
- **Verification procedures** for independent auditors
- **Conformance tests** enabling cross-implementation interoperability

As autonomous systems gain authority in financial, infrastructure, healthcare, and government environments, verifiable decision records will become a foundational requirement for trustworthy AI systems.

The Capsule Protocol Specification (CPS) is offered as a concrete reference design demonstrating how such a system can be implemented in practice. Our goal is not to establish a proprietary technology, but to contribute a working model that can inform future standards development for accountable AI systems.

CPS is intentionally storage-agnostic, vendor-neutral, and independent of any specific AI framework.

> **How to Evaluate This Submission**
>
> Clone the repository at `github.com/quantumpipes/capsule`. Run the conformance tests in either language (`pytest` or `npm test`). Create a Capsule. Seal it. Tamper with it. Watch verification fail. The protocol is designed to be verifiable, and we encourage NIST to verify it.

THE CORE PRINCIPLE

No autonomous AI action should occur without a structured, cryptographically sealed record of what happened and why.

`∀ action : ∃ capsule`

References

Capsule Protocol Specification (CPS) v1.0

Complete protocol specification including record schema, cryptographic sealing and verification procedures, hash chain rules, canonical serialization, and golden test vectors for cross-implementation conformance. Available at `github.com/quantumpipes/capsule`.

Open-Source Reference Implementations

`github.com/quantumpipes/capsule` (Apache-2.0 with perpetual, royalty-free patent grant).
Python and TypeScript reference implementations, both conformant (15/15 golden fixtures), 600+ tests combined with 100% coverage in both languages. Protocol-first repository: `spec/`, `conformance/`, `reference/`.

NIST Publications Referenced

SP 800-53 Rev. 5 · AI RMF 1.0 (AI 100-1) · SP 800-218A · AI 100-2 (2025) · AI 600-1 · SP 800-207 (Zero Trust Architecture)

Additional Sources

OWASP Top 10 for Agentic Applications (December 2025) · NCCoE Concept Paper: Accelerating the Adoption of Software and AI Agent Identity and Authorization (February 2026) · CyberArk 2025 Identity Security Landscape

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Open-source trust infrastructure for autonomous AI.

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